University of Caloocan City

Computer Studies Department

CCS 119 – Capstone Project

EduPredict: Questionnaire Scoring, Computation, and Supporting Studies

Updated 60-Item Scoring and Research Basis (Consultation Draft)

Calma, Keith Andrea S. • Asistio, Chiel C.

Esquillo, Xyra Jeanne V. • Orosco, Jana Madelaine T.

October 2026

1. Purpose and Status

This document updates the original EduPredict scoring and research-basis paper from 36 to 60 questionnaire items across six sections. It explains the questionnaire computations, the scientific background of the constructs, the intended roles of each section in the system, and the procedures for validating both the instrument and the predictive models. It is designed for consultation and as a technical reference during ML development.

The 60-item structure reflects the latest panel instruction of ten questions per section. Panel-requested structure is not the same as psychometric validation. The revised items still require expert review and pilot testing. No expert ratings, reliability values, or completed ML performance results are reported here. The exact wording and numbering must be checked against the latest approved-questionnaire file before software implementation.

2. Instrument Overview and System Roles

Every item uses a five-point agreement response. Questions apply broadly to CLAS students, regardless of degree program. Student grades, structured skills, certifications and experience may also be available through other EduPredict modules; these are separate records, not automatically equivalent to self-report questionnaire responses.

Code	Construct	Items	Role in EduPredict
A	Academic factors / academic self-perception	A1–A10	Descriptive academic context; compared with actual grades. Not a trained ML input.
S	Socioeconomic factors / learning resources	S1–S10	Descriptive financial and access context; support recommendations. Not a trained ML input.
B	Behavioral factors / student engagement	B1–B10	Separate engagement assessment and rule-based support. Does not modify dropout model probability.
C	Competence / employability skills	C1–C10	Candidate source-compatible predictor for employability regression, subject to construct mapping and validation.
P	Psychological capital / career readiness	P1–P10	Candidate source-compatible predictor for employability regression, subject to construct mapping and validation.

Code	Construct	Items	Role in EduPredict
E	Self-perceived employability	E1–E10	Separate descriptive self-assessment; compare with modeled readiness. Never feed as predictor for employability target.

The dropout model is separately trained with compatible enrollment, academic, and other verified profile features from the UCI source data. A, S and B questionnaire means must not be disguised as the source dataset's dropout predictors.

3. Questionnaire Scoring and Computation

3.1 Five-point response scale

Numeric response	Verbal response
1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

The latest 60-item draft is positively worded; therefore, no reverse-coded UCC items are specified at present. Reverse coding in any original external dataset, however, must be audited independently. A missing response is not automatically assigned 3 (Neutral). An incomplete section should remain incomplete until the survey completion policy is explicitly defined.

3.2 Mean score for each independent section

Construct mean = Sum of the ten item responses / 10

AcademicMean = (A1 + A2 + … + A10) / 10

SocioeconomicMean = (S1 + S2 + … + S10) / 10

BehavioralMean = (B1 + B2 + … + B10) / 10

CompetenceMean = (C1 + C2 + … + C10) / 10

PsyCapMean = (P1 + P2 + … + P10) / 10

SelfPerceivedEmployabilityMean = (E1 + E2 + … + E10) / 10

Each mean is on the original 1.00–5.00 scale. Section scores should be reported independently; adding all 60 questions into one total would mix different constructs. Mean-based analysis of multi-item Likert scales should be interpreted with appropriate assumptions (Boone & Boone, 2012).

3.3 Five descriptive interpretation levels

The accepted consultation draft divided the interval from 1 to 5 into five equal widths: (5 − 1) / 5 = 0.80 (Pimentel, 2010). The following are descriptive conventions, not empirically calibrated dropout-risk or hiring-probability categories.

Mean interval	Interpretation	Equivalent rescaled index	A/S/B context flag
1.00–1.80	Very Low	0–20	Yes
Above 1.80–2.60	Low	Above 20–40	Yes
Above 2.60–3.40	Moderate	Above 40–60	No
Above 3.40–4.20	High	Above 60–80	No
Above 4.20–5.00	Very High	Above 80–100	No

For presentation, conventional two-decimal labels 1.81–2.60, 2.61–3.40, 3.41–4.20 and 4.21–5.00 can be used. In code, compare unrounded means with boundaries at 1.80, 2.60, 3.40 and 4.20 to avoid rounding errors. An A, S or B mean at or below 2.60 is a proposed contextual support flag, not an ML model prediction or validated clinical/educational-risk threshold. A high B mean indicates high engagement, not high risk.

3.4 Employability Readiness Index

The planned employability model is a regressor trained on Pianda et al. (2026) source-data constructs. Its candidate deployment inputs are Competence and Psychological Capital scores, only after a documented review determines whether EduPredict's 10-item C/P means can validly represent the original constructs. The original target (EVS/employability) must be scored using its authentic source codebook; source target items must not be leaked into predictors.

If the verified target uses a 1–5 native range, the predicted score can be linearly rescaled for display:

$$\textit{Employability Readiness Index} = ((\textit{predicted score} - 1) / 4) \times 100$$

This is a display index from 0 to 100, not the probability of finding a job. Store the original model score, displayed index, feature/schema version, dataset/model version and timestamp. Do not claim that questionnaire mean scores are directly interchangeable with Pianda's constructs until evidence supports that mapping.

3.5 Updated worked example (illustrative only)

Section	Ten example answers	Sum	Mean	Level
C	4, 5, 4, 3, 4, 4, 5, 4, 4, 4	41	4.10	High
P	4, 4, 5, 4, 3, 4, 4, 5, 4, 4	41	4.10	High
E	4, 4, 3, 4, 4, 3, 4, 4, 4, 4	38	3.80	High
B	2, 3, 2, 4, 2, 4, 3, 3, 4, 2	29	2.90	Moderate

Suppose a hypothetical model predicts native employability 3.90. The displayed index is ((3.90 − 1)/4) × 100 = 72.5/100 (High). The student's E mean of 3.80 independently corresponds to 70/100. Their 2.5-point index difference can be presented for discussion, not as evidence of prediction accuracy. The example is invented solely to demonstrate arithmetic; no model output is being claimed.

3.6 Contextual recommendations and separation of results

Low A/S/B descriptive scores may be linked to student-support suggestions (academic advising, financial assistance, study resources, attendance, time management or instructor consultation). These rules should be clearly

marked as questionnaire-based observations. Dropout model factors and the estimated probability of the Dropout class should be shown separately. Questionnaire behavior cannot change a dropout probability merely because a recommendation rule fired.

4. Validation of the Instrument

4.1 Expert content validation

Experts should assess each of the 60 items for relevance, clarity and appropriateness to UCC CLAS using a four-point review scale. Compute, for each item, the item-level content-validity index (I-CVI) and then the scale average (S-CVI/Ave):

$$I\text{-}CVI = \text{Experts scoring the item 3 or 4} / \text{Number of experts}$$

$$S\text{-}CVI/Ave = \text{Mean of item I-CVIs}$$

The previously cited literature discusses reference values such as I-CVI ≥ 0.78 and S-CVI/Ave ≥ 0.90 (Lynn, 1986; Polit & Beck, 2006), which must be interpreted in light of the actual number of experts and review design. Track all proposed wording changes. No observed CVI value is available yet.

4.2 Pilot testing and construct reliability

Pilot with suitable CLAS-like students, then calculate Cronbach's alpha separately for each of A, S, B, C, P and E, using k = 10 items per section:

$$\alpha = [k / (k - 1)] \times [1 - (\text{sum of item variances} / \text{variance of summed item scores})]$$

A conventional alpha threshold around 0.70 may guide discussion, but it does not establish unidimensionality or content validity. Inspect item behavior and potential factor overlap; do not remove items purely to improve alpha. Reliability and transferability of C/P scores to the external employability scale require additional evidence. A questionnaire pilot without real later dropout or employment outcomes cannot validate predictions.

5. Model Development and Evaluation

5.1 Employability readiness: Pianda et al. regression

The selected external source is Pianda et al. (2026), Mendeley Data Version 4 (DOI: 10.17632/hmn3b4c6c4.4), linked to the Data in Brief article. It contains approximately 266 vocational students from Aceh, Indonesia and constructs related to internship experience, competence, psychological capital and employability. The original item counts, reverse-coding rules and exact target scoring must be verified from the real dataset and codebook before training.

Candidate deployment-compatible predictors: original source Competence and Psychological Capital construct scores; target: original source Employability construct score. Consider an expanded source-native internship model only as a clearly labeled exploratory comparison, because internship responses may not be consistently available at deployment. Use a baseline and candidates such as Linear Regression, Ridge, Random Forest and Gradient Boosting; prevent preprocessing leakage through training-fold pipelines and report repeated cross-validation variability.

$$MAE = (1/n) \times \text{sum } |\text{actual} - \text{predicted}|$$

$$RMSE = \text{square root of } [(1/n) \times \text{sum (actual} - \text{predicted)}^2]$$

$$R^2 = 1 - (\text{residual sum of squares} / \text{total sum of squares})$$

Regression accuracy percentages are not appropriate metrics. Compare candidate models on out-of-sample MAE, RMSE, R^2, variance across folds and overfitting. Do not publish invented values, or suggest source performance confirms applicability to UCC.

5.2 Student dropout: UCI multi-class classification

The locked source is UCI's Predict Students' Dropout and Academic Success (Realinho et al.; DOI: 10.24432/C5MC89), approximately 4,424 records with three outcome labels: Dropout, Enrolled, Graduate. These are source outcome classes, not Low/Moderate/High risk bands. Candidate classifiers may include baseline, Logistic Regression, Random Forest and Gradient Boosting models. Preserve a stratified evaluation design and select features that can truly be obtained from EduPredict at the intended prediction time.

Evaluate accuracy, class-wise precision/recall/F1, macro F1, confusion matrix, and appropriate multi-class ROC-AUC. Prioritize sensitivity to the Dropout class, check probability calibration, and explicitly report which source variables were excluded. If a probability for the Dropout class is used, its Low/Moderate/High presentation thresholds require validation and justification; no arbitrary one-third cutoffs.

Do not send UCC 1.00-best/5.00-fail grades directly as if they were the Portuguese source's 0–20 grades. Any defensible transformation must match the educational meaning and timing of records, including semester status and course-unit counts. Unavailable source variables must not be filled with made-up UCC data. Behavioral B responses remain a separate descriptive score, not a hidden dropout-model input.

5.3 Reproducibility and responsible deployment

For both models, record source-data versions and provenance, source feature schema, preprocessing rules, cross-validation setup, random seeds, candidate scores, chosen pipeline and limitations. Save an inference artifact only after testing. Validate input schemas before any later Laravel-to-FastAPI connection. The current heuristic predictor must not be described as a trained research model.

6. Supporting Studies for the Questionnaire and Results

6.1 Academic factors (A1–A10)

Bandura (1997) provides the self-efficacy foundation, while Chemers, Hu and Garcia (2001) connect academic self-efficacy with first-year college adjustment and performance. Tinto (1975) provides a broader framework linking academic integration and student continuation. A1–A10 cover satisfaction with performance, lesson comprehension, workload management, effort, passing courses, completing the degree, explaining learning, assessment confidence, improving grades and identifying areas for improvement. These are contextual self-perceptions, not substitutes for recorded grades or directly tested dropout predictors.

6.2 Socioeconomic factors (S1–S10)

Northern, O'Brien and Goetz (2010) developed a financial stress scale for undergraduate students, supporting investigation of strain and educational resources. S1–S10 cover budgets, transport, access to devices and internet, ability to focus, continuation support, unexpected expenses, meals, study space and competing household/work obligations. These are context-specific EduPredict adaptations and not validated direct copies of the cited scale; expert review must assess suitability and potential sensitive-information concerns.

6.3 Behavioral factors (B1–B10)

Fredricks, Blumenfeld and Paris (2004) describe behavioral, emotional and cognitive dimensions of student engagement. Maroco et al. (2016) developed the University Student Engagement Inventory (USEI), providing a university-level conceptual reference. B1–B10 focus on attendance, study routines, class participation, learning resources, deadlines, persistence, note-taking, study groups, seeking help and maintaining focus. The ten EduPredict questions are not the original USEI; do not claim an official USEI-derived score or a separately trained behavioral ML model.

6.4 Competence (C1–C10)

Pianda et al. (2026) provide the selected external dataset linking competence with employability-related survey measures; Van der Heijde and Van der Heijden (2006) frame employability as multidimensional competence. Fugate, Kinicki and Ashforth (2004) emphasize psychosocial employability resources. C1–C10 cover communication, teamwork, problem solving, adaptability, digital tools, applying lessons, organization, skill learning, professional responsibility and accepting feedback. Similar labels do not establish exact item/scale equivalence to the original Pianda training features.

6.5 Psychological capital (P1–P10)

Luthans, Avolio, Avey and Norman (2007) describe psychological capital through efficacy, hope, optimism and resilience. Xiong et al. (2021) report a relationship between psychological capital and employability in a vocational-student setting. P1–P10 cover confidence with difficulties, alternative solutions, persistence, optimism, recovery after setbacks, growth belief, multiple pathways, motivation, adjusting plans and a positive outlook. The adapted ten-item score remains to be locally validated and compared carefully against the original Pianda construct.

6.6 Self-perceived employability (E1–E10)

Rothwell, Herbert and Rothwell (2008) developed a self-perceived employability scale for university students. E1–E10 cover perceived preparation, demonstrating skills, explaining experience, career learning, job opportunity beliefs, market awareness, employer valuation, interview confidence, alternative options and planning after graduation. This section reflects the student’s own appraisal; it should not enter the employability regression as an input because of conceptual circularity with the predicted target.

6.7 Research supporting dropout modeling

Realinho et al. (2022) document the underlying higher-education dropout and academic-success data. Kim, Yoo and Kim (2023) examine machine-learning prediction and feature relevance using a related subset; Moreira da Silva et al. (2022) provide another Portuguese university dropout modeling example. Alyahyan and Düştegör (2020) review higher-education prediction methods and practices. Published findings motivate model comparison; any numeric performance reported in those papers belongs to the authors’ experiments, not to EduPredict.

7. Limitations, Ethics and Next Research Steps

The source employability respondents were vocational students in Indonesia, while the intended users are UCC CLAS students across multiple programs. Source constructs, wording, educational environments and measurement properties may differ, even if both use five-point scales. The source dataset is small, and self-perceived readiness is neither hiring probability nor a verified labor-market outcome. The 60-item instrument extends previously drafted items; independent expert judgment, pilot psychometrics and local validation are necessary before strong validity claims.

The dropout dataset reflects Portuguese higher education. Its admission, economic and academic features may not match UCC's fields, grading rules or collection points. Any proposed classifier should be tested on a defensible deployment-compatible feature subset rather than treated as production-ready because a full-feature external benchmark scores well. A model-based risk label is a support tool, not a decision to exclude or penalize students.

Follow Philippine Data Privacy Act (RA 10173) principles through appropriate consent/notice, access restrictions, data minimization, secure retention and careful handling of student data. Future work should seek real UCC program-appropriate outcomes to assess local performance, calibration, fairness and drift. Recommendations should remain supportive and subject to faculty review.

8. Pre-Training Reference Checklist

• Confirm that the separate EduPredict_Questionnaire.pdf is the current exact 60-item, ten-per-section instrument; verify codes and text.

• Provide this updated basis to Cursor; ensure all six construct means use ten valid responses and no all-60 total is calculated.

• Audit Pianda v4 raw columns, original codebook, target scoring, reverse coding and C/P mapping; never substitute invented source values.

• Audit UCI target class labels, source features, grade-scale semantics and which records are available in EduPredict at prediction time.

• Train and evaluate only source-grounded candidates; preserve results and clearly disclose unknown local transfer validity.

• Do not update production interfaces or assert successful FastAPI deployment until model and integration tests exist.

References

Alyahyan, E., & Düştegör, D. (2020). Predicting academic success in higher education: Literature review and best practices. International Journal of Educational Technology in Higher Education, 17, Article 3.

Bandura, A. (1997). Self-efficacy: The exercise of control. W. H. Freeman.

Boone, H. N., & Boone, D. A. (2012). Analyzing Likert data. Journal of Extension, 50(2).

Chemers, M. M., Hu, L., & Garcia, B. F. (2001). Academic self-efficacy and first-year college student performance and adjustment. Journal of Educational Psychology, 93(1), 55–64.

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. Psychometrika, 16(3), 297–334.

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. Review of Educational Research, 74(1), 59–109.

Fugate, M., Kinicki, A. J., & Ashforth, B. E. (2004). Employability: A psycho-social construct, its dimensions, and applications. Journal of Vocational Behavior, 65(1), 14–38.

Kim, S., Yoo, E., & Kim, S. (2023). Why do students drop out? University dropout prediction and associated factor analysis using machine learning techniques [Preprint]. arXiv:2310.10987. https://arxiv.org/abs/2310.10987

Luthans, F., Avolio, B. J., Avey, J. B., & Norman, S. M. (2007). Positive psychological capital: Measurement and relationship with performance and satisfaction. Personnel Psychology, 60(3), 541–572.

Lynn, M. R. (1986). Determination and quantification of content validity. Nursing Research, 35(6), 382–385.

Maroco, J., Maroco, A. L., Campos, J. A. D. B., & Fredricks, J. A. (2016). University student's engagement: Development of the University Student Engagement Inventory (USEI). Psicologia: Reflexão e Crítica, 29, Article 21.

Moreira da Silva, D. E., Solteiro Pires, E. J., Reis, A., Moura Oliveira, P. B., & Barroso, J. (2022). Forecasting students dropout: A UTAD University study. Future Internet, 14(3), 76. https://doi.org/10.3390/fi14030076

Northern, J. J., O'Brien, W. H., & Goetz, P. W. (2010). The development, evaluation, and validation of a financial stress scale for undergraduate students. Journal of College Student Development, 51(1), 79–92.

Nunnally, J. C., & Bernstein, I. H. (1994). Psychometric theory (3rd ed.). McGraw-Hill.

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2024). The impact of internship experience on the employability of vocational students: A bibliometric and systematic review. Cogent Business & Management, 11(1), 2386465.

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2026). A stratified survey dataset on internship experience, competencies, psychological capital, and employability of vocational students in Aceh, Indonesia. Data in Brief, 67, 112992. https://doi.org/10.1016/j.dib.2026.112992

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2026). Dataset of student internship experience, competence, psychological capital and employability's vocational student (Version 4) [Data set]. Mendeley Data. https://doi.org/10.17632/hmn3b4c6c4.4

Pimentel, J. L. (2010). A note on the usage of Likert scaling for research data analysis. USM R&D Journal, 18(2), 109–112.

Polit, D. F., & Beck, C. T. (2006). The content validity index: Are you sure you know what's being reported? Critique and recommendations. Research in Nursing & Health, 29(5), 489–497.

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2022). Predicting student dropout and academic success. Data, 7(11), 146.

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict students' dropout and academic success [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89

Rothwell, A., Herbert, I., & Rothwell, F. (2008). Self-perceived employability: Construction and initial validation of a scale for university students. Journal of Vocational Behavior, 73(1), 1–12.

Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. Review of Educational Research, 45(1), 89–125.

Van der Heijde, C. M., & Van der Heijden, B. I. J. M. (2006). A competence-based and multidimensional operationalization and measurement of employability. Human Resource Management, 45(3), 449–476.

Xiong, L., Zhang, Q., Cao, Y., Yang, J., & Yang, M. (2021). An exploration into the influence of higher vocational students' psychological capital on their employability. SHS Web of Conferences, 96, 02007. https://doi.org/10.1051/shsconf/20219602007

Reference note: Bibliographic entries are retained from the previous consultation paper; source-article details and the external dataset codebooks should be independently checked when finalizing the capstone manuscript.